

DAA Assignment #1

Date

24K-0013

Q1

Insertion Sort : initial array = [18, 7, 25, 14, 3, 21, 10, 30]

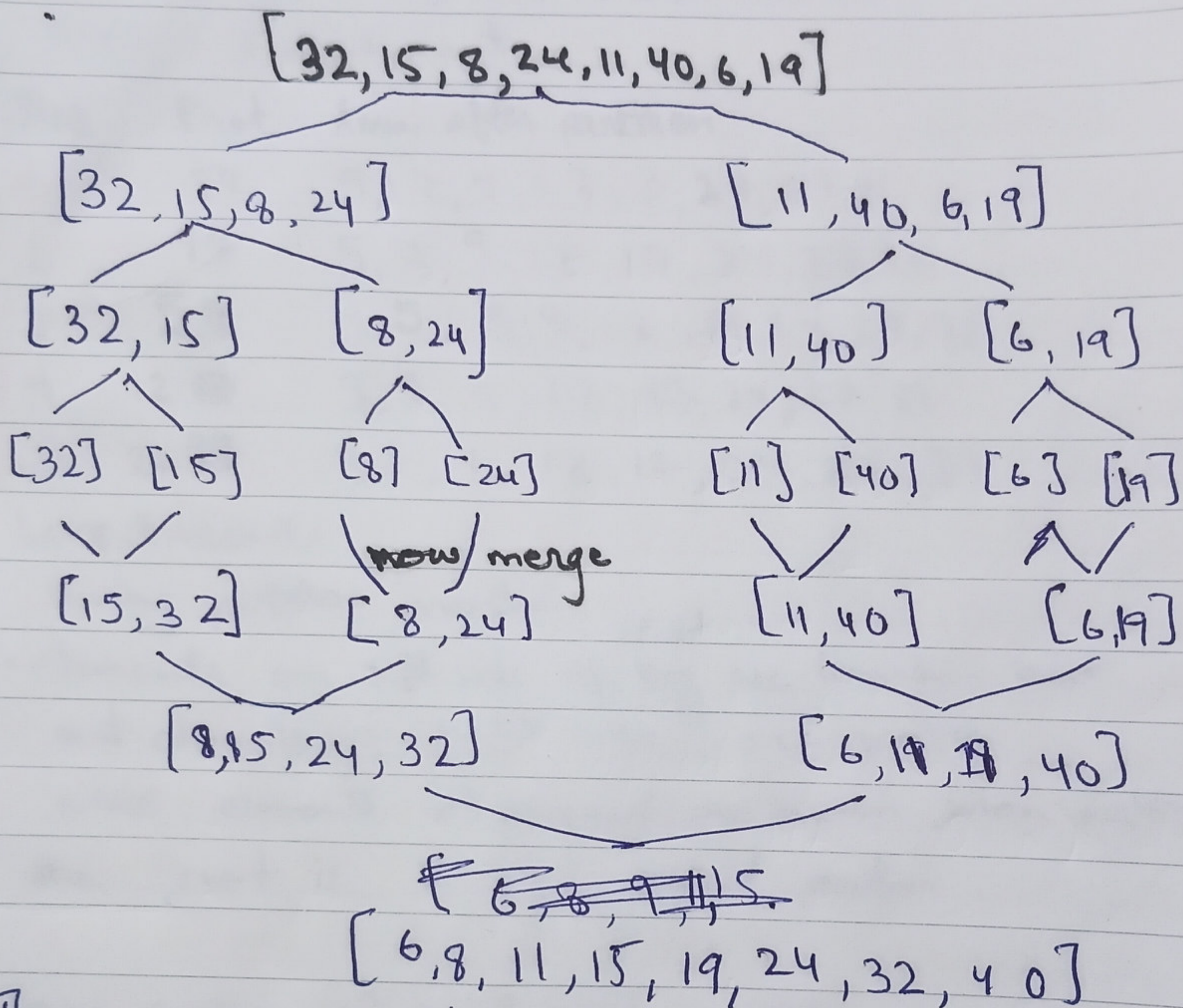
iteration	key	Array after insertion.
1	7	7, 18, 25, 14, 3, 21, 10, 30
2	25	7, 18, 25, 14, 3, 21, 10, 30
3	14	7, 14, 18, 25, 3, 21, 10, 30
4	3	3, 7, 14, 18, 25, 21, 10, 30
5	21	3, 7, 14, 18, 21, 25, 10, 30
6	10	3, 7, 10, 14, 18, 21, 25, 30
7	30	3, 7, 10, 14, 18, 21, 25, 30

In worst case, every key is compared to all its previous elements
 $1 + 2 + 3 + \dots + (n-1) = n(n-1)/2 = O(n^2)$

Invariant : before each iteration, the elements before the key are sorted.

- Initialization: One element is already sorted.
- Maintenance: Key is inserted into its correct position.
- Termination: All elements are sorted.

Q#2



Three way merge sort: $T(n) = 3T(n/3) + O(n)$
 using Master theorem: $a=3, b=3, k=1 = \log_b a = 1$
 $T(n) = O(n \log n)$

Q3

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initial array = [24, 9, 17, 5, 31, 12, 27, 3]
 Pivot = First element

Step	Pivot	Array after partition
1	24	9, 17, 5, 12, 3, 24, 27, 31
2	9	5, 3, 9, 12, 17, 24, 27, 31
3	5	3, 5, 9, 12, 17, 24, 27, 31
4	12	3, 5, 9, 12, 17, 24, 27, 31
5	27	3, 5, 9, 12, 17, 24, 27, 31

Loop Invariant:

- During partition maintain
- elements on left side of ^(Pivot) key are less than pivot.
 and elements of Right side are greater.

while elements not processed are known, when partition ends
 the pivot is in its correct position.

Q4) For pairing we select x_i and x_j ~~points~~ elements points for i & j teams, and we ~~interpret~~ ~~select~~ i & j for each player selection from a sorted array in $O(n)$ time.

$O(n^2)$: Use bubble sort to sort the array then make pairs or teams.

$O(n \log n)$: Use merge sort to sort the array, then make pairs or teams.

Q5 Pairing: combine i^{th} smallest with i largest:

Pair 1: $(2, 12) \Rightarrow \text{sum} = 14$, Pair 2: $(4, 10) \Rightarrow \text{sum} = 14$, Pair 3: $(6, 8) \Rightarrow \text{sum} = 14$

Pair sums = 14, 14, 14; max pair sum = 14

Algorithm: sorting $O(n \log n)$ and pair $O(n)$

$\downarrow$
use merge sort

$\downarrow$
combine x_i with x_{n-i+1} term.

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Q 6

$$n^2 + 5n + 10 = O(n^2)$$

$$\lim_{n \rightarrow \infty} \frac{n^2 + 5n + 10}{n^2} \leq c$$

$$\frac{n^2}{n^2} + \frac{5n}{n \cdot n} + \frac{10}{n^2} \leq c$$

$$\boxed{c \geq 1}$$

So, $c = 2$ & $n_0 = 10$ holds

$$4n^2 + \log_2 n + 3n^2 = O(n^2 \log_2 n)$$

$$\lim_{n \rightarrow \infty} \frac{4n^2}{n^2 \log_2 n} + \frac{\log_2 n}{n^2 \log_2 n} + \frac{3n^2}{n^2 \log_2 n} \leq c$$

$$\lim_{n \rightarrow \infty} (4 + 3/\log n) \leq c$$

$$c \geq 0 + 4 \Rightarrow c \geq 4$$

$$10^2 + 5(10) + 10 \leq 2(10)^2$$

$$160 \leq 200$$

$$\left. \begin{array}{l} \text{take } c = 7, n_0 = 2 \end{array} \right\}$$

$$4(2)^2 + 3(2)^2 \leq 7(2)^2$$

$$28 \leq 28$$

$$\boxed{c = 7} \quad \boxed{n_0 = 2}$$

Q 7

Asymptotic notation describes how an algorithm's running time & space grows as input size n increases, ignoring machine-dependent

Asymptotic notation describes how an algorithm's running time & space grows as input size n increases, ignoring machine-dependent details.

Big-O: Gives upper bound on growth

Big- Ω : Gives lower bound on growth

Big Θ : Gives tight bound.

ex#

$$f(n) = 3n^2 + 5n + 2$$

$$f(n) = O(n^2), f(n) = \Omega(n^2), f(n) = \Theta(n^2)$$

For every larger n , the highest order^{term} of n dominates the growth rate, so we neglect lower order terms.

$$\text{Algo A: } O(n^2) \quad \left\{ \Rightarrow A \approx 10000 \right.$$

$$\text{Algo B: } O(n) \quad \left\{ \Rightarrow B \approx 100 \right.$$

For $n = 100$ -

Thus $O(n)$ grows much slower than $O(n^2)$ as input size increases

Q8

$$T(n) = \sqrt{2} T(n/2) + \log n$$

$$a = \sqrt{2}, b = 2, k = 0$$

$$\log_b a = 1/2 > k$$

$$: \text{Case \#1} \rightarrow \Theta(n^{\log_b a})$$

$$f(n) = \Theta(\sqrt{n})$$

$$T(n) = aT(n/b) + f(n)$$

$$f(n) = n^k \log^p n$$

$$\Theta(n^{\log_b a})$$

$$T(n) = 64 T(n/8) - n^2 \log n$$

$$a = 64, b = 8, f(n) = -n^2 \log n$$

since, $f(n)$ is not positive, master theorem cannot be applied.

$$T(n) = 16 T(n/4) + n!$$

$$a = 16, b = 4, \log_b a = 2, f(n) = n!$$

$$f(n) > n^{\log_b a}$$

$$\text{Case \#3: } a f(n/b) \leq c \cdot f(n) \Rightarrow 16 \left(\frac{n}{4}\right)! \leq c \cdot n!$$

For larger n , holds true for $c \leq 1$

$$T(n) = \Theta(n!)$$

Q9)

For pairing we select n and $n-1$ and $n-2$ and $n-3$ and so on.

Q9) (i) $T(n) = 2T(n/2) + n \log n$ (ii) $T(n/2) = 2T(n/4) + \frac{n}{2} \log \frac{n}{2}$
 (iii) $T(n/4) = 2T(n/8) + \frac{n}{4} \log \frac{n}{4}$

Substitute eq (ii) & (iii) in (i)

$$T(n) = 4T(n/4) + n \log_2 n/2 + n \log_2 n$$

$$T(n) = 8T(n/8) + 3n \log_2 n - 3n$$

Continue till k steps

$$T(n) = 2^k T(n/2^k) + kn \log_2 n - n(k(k-1))/2$$

$$n/2^k = 1 \rightarrow k = \log_2 n$$

$$T(n) = 2^{\log_2 n} T(1) + (\log_2 n) n \log_2 n - n(\log_2 n)(\log_2 n - 1)/2$$

$$\boxed{T(n) = \Theta(n \log n)^2}$$

$$T(n) = 8T(n/2) + n^2 \quad (i) \quad T(n/2) = 8T(n/4) + (n/2)^2 \quad (ii)$$

Substitute eq (ii) & (iii) in eq (i) $T(n/4) = 8T(n/8) + (n/4)^2$ (iii)

$$T(n) = 8[8T(n/4) + n^2/4] + n^2$$

$$T(n) = 512T(n/8) + 4n^2 + 3n^2$$

$$T(n) = 8^k T(n/2^k) + n^2(2^k - 1)$$

$$= 8^k T(1) + n^2(2^{\log_2 n} - 1)$$

$$= 8^k + n^2(n - 1)$$

$$= 8^k + n^3 - n^2$$

$$\boxed{T(n) = \Theta(n^3)}$$

$\therefore$ Assume $n/2^k = 1$

$$k \Rightarrow \log_2 n$$

$$8^k \Rightarrow n^3$$

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~~Q10~~

Q10) (a) True, eg: $n^2 + O(n) = \Theta(n)^2$; $2^2 + O(2) = \Theta(2^2)$

$$b \leq 4$$

(b) True, transitivity, $\lim_{n \rightarrow \infty} \left(\frac{f(n)}{g(n)} \right) = c_1$ & $\lim_{n \rightarrow \infty} \frac{g(n)}{h(n)} \Rightarrow c_2$
then $\lim_{n \rightarrow \infty} \frac{f(n)}{h(n)} = c_1 c_2 \Rightarrow f(n) = O(h(n))$

(c) True, since $f(n) = \Theta(g(n))$, $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = c$, thus $f(n) + g(n) = \Theta(g(n))$

(d) True, $\lim_{n \rightarrow \infty} \frac{n^2}{n^3} = 0 \leq c$, Holds true for $c \neq 1$ & $n_0 = 1$

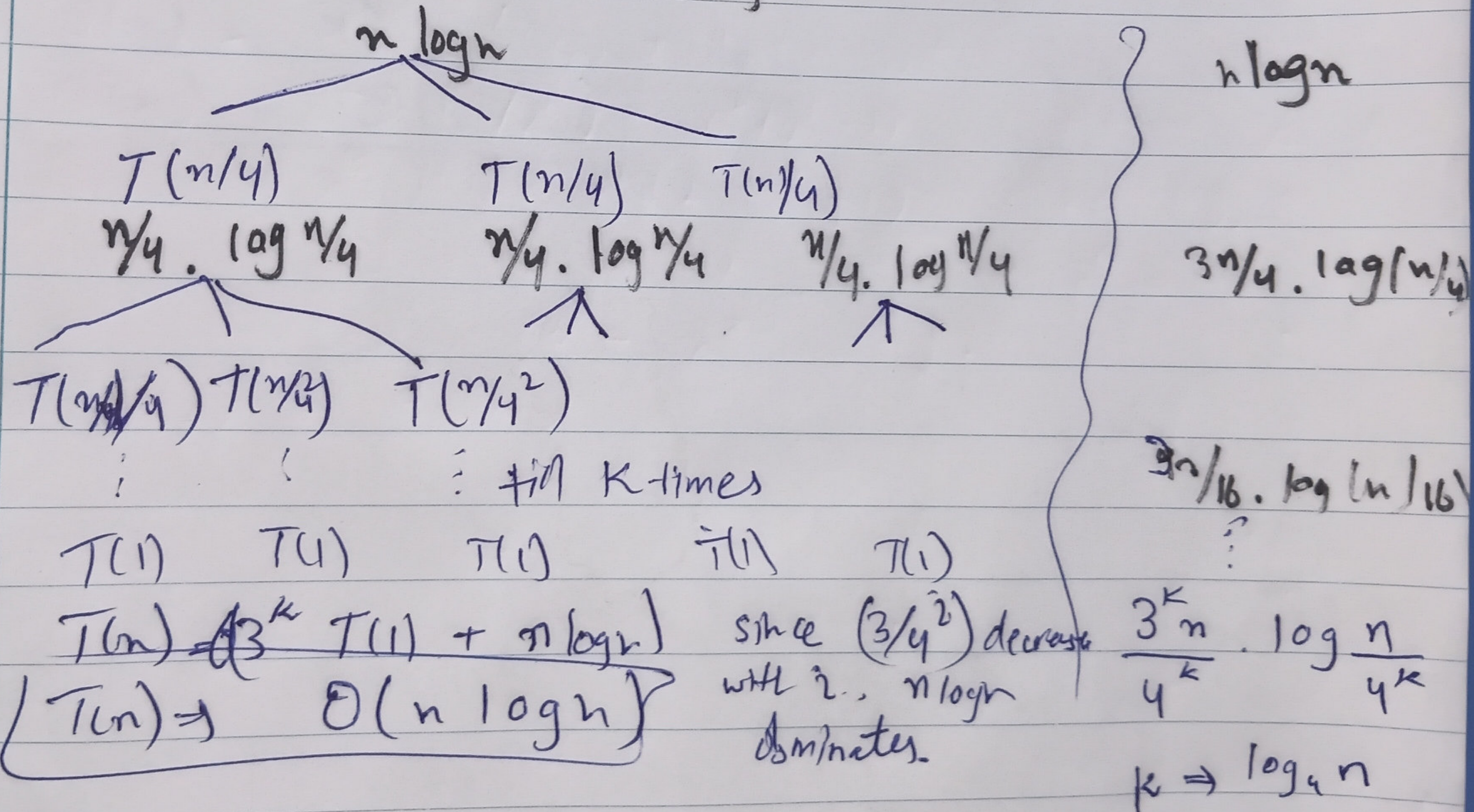
(e) False, $\lim_{n \rightarrow \infty} \frac{n^3}{n^2} = \infty$. No constant value can bound it from above

Q11) a $T(n) = 3T(n/4) + n \log n$

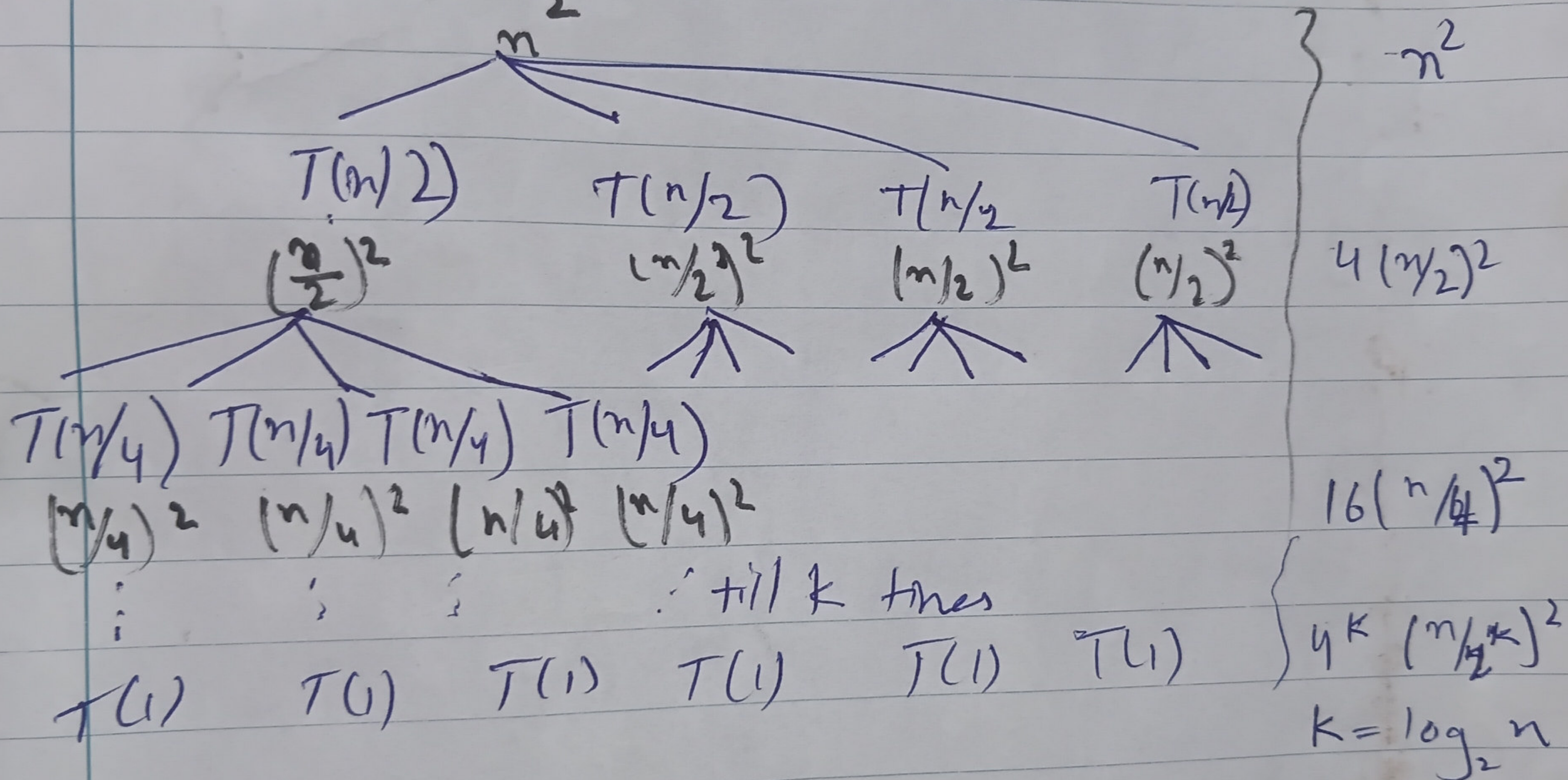
? $n \log n$

⑥ False, $\lim_{n \rightarrow \infty} \frac{n^3}{n^2} = \infty$. No constant value can bound it from above

Q11) a $T(n) = 3T(n/4) + n \log n$



Q11)(b) $T(n) = 4T(n/2) + n^2$



$$T(n) = 4^k T(1) + n^2 \log n$$

$$T(n) = \Theta(n^2 \log n)$$

$n \cdot \log n$ is the sum of cost of each node